Group II, Row IV

(checked; corrections and required clarifications marked in red)

Summary of the check. The derivation is correct: the prefactor of the $\mathcal{P}_{gg}$ form is right ($\frac{g^4}{16\pi^4}$), the soft channel is counted exactly once, and the add-and-subtract of the $\xi \to 0$ pole in the δ-function representation is exact. Three points needed fixing or tightening: (1) the *final* JIMWLK category had a $-$ sign, inconsistent with the (correct) $+$ of the same equation derived earlier in this note; (2) the plus prescription $\left[\frac{1-\xi}{\xi}\right]_+$ is well-defined only in the δ/w-representation — carrying it back under the -bracket and into $\mathcal{P}_{gg}$ is a *shorthand* that must be declared, because the subtraction piece is not of the form $\times$(anything); the defining equation is now stated. Relatedly, this $\mathcal{P}_{gg}$ is *not* the textbook splitting function (which has an unregulated $\frac{1-\xi}{\xi}$) — the difference is precisely the BFKL subtraction. (3) The "Finite terms" block still contains the endpoint $\log(1-\frac{k^+}{\vee})$ entry, which carries $\frac{2}{\epsilon}$; a remark now says this is the plus-prescription convention term at $\xi = 1$ and how to absorb it if a truly finite block is wanted. Here $\equiv \frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}$.

1 The derivation

The setup for the cross section:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{4}{(2\pi)^3}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y,y'}\Big[&\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+ + k^+, -y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+, y'-w'; p^+, w'-z)\\ &- \bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+ + k^+, -y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+, y'-w'; p^+, w'-z)U^{d'b}(z)\Big]\times\Big[\bar{\mathbb{C}}^{dac}_{1jik}(p^+ + k^+, y-w; p^+, w-z)\\ &U^{db}(z)U^{ce}(y)\mathbb{A}^{(1)\,e}_k(p^+ + k^+, -y) - U^{db}(z)\bar{\mathbb{C}}^{dac}_{1jik}(p^+ + k^+, y-w; p^+, w-z)\bar{\mathbb{A}}^{(1)\,c}_k(p^+ + k^+, -y)\Big]\end{aligned}$$

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w',x'} e^{-ik(w'-w)}\int_{x,y,y'}\frac{(w-z)^m\,(x-y)^k\,(w'-z)^{m'}\,(x'-y')^{k'}}{(w-z)^2\,(x-y)^2\,(w'-z)^2\,(x'-y')^2}\,\frac{p^+\,k^+}{(p^+ + k^+)^4}\\ \delta^{(2)}\left[y - w + \frac{p^+}{(p^+ + k^+)}\,(w-z)\right]\times\delta^{(2)}\left[y' - w' + \frac{p^+}{(p^+ + k^+)}\,(w'-z)\right]\times\\ \left[\delta_{km}\,\delta_{ij} - \frac{(p^+ + k^+)}{p^+}\,\delta_{jm}\,\delta_{ik} - \frac{(p^+ + k^+)}{k^+}\,\delta_{im}\,\delta_{kj}\right]\times\left[\delta_{k'm'}\,\delta_{ij} - \frac{(p^+ + k^+)}{p^+}\,\delta_{jm'}\,\delta_{ik'} - \frac{(p^+ + k^+)}{k^+}\,\delta_{im'}\,\delta_{k'j}\right]\\ \left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cda}\,U^{ce}(y)\,U^{db}(z)\,\rho^{e}(x) - f^{cda}\,U^{db}(z)\,U^{ce}(x)\,\rho^{e}(x)\right]\end{aligned} \tag{1.1}$$

Integrating w and w',

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_z e^{-ik\,(y'-y)\,\frac{(k^+ + p^+)}{k^+}}\int_{x,y,y',x'}\frac{(y-z)^m\,(x-y)^k\,(y'-z)^{m'}\,(x'-y')^{k'}}{(y-z)^2\,(x-y)^2\,(y'-z)^2\,(x'-y')^2}\\ \left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^+ + p^+)^2} + \frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k} - \delta_{m'k'}\delta_{km}) + \frac{p^+}{k^{+\,2}}\,\delta_{mm'}\delta_{kk'} + \frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right]\times\\ \left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cda}\,U^{ce}(y)\,U^{db}(z)\,\rho^{e}(x) - f^{cda}\,U^{db}(z)\,U^{ce}(x)\,\rho^{e}(x)\right]\end{aligned} \tag{1.2}$$

Reducing Wilson lines, we get

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\frac{N_c}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_z e^{-ik\,(y'-y)}\,e^{-ik\,(y'-y)\,\frac{p^+}{k^+}}\int_{x,y,y',x'}\frac{(y-z)^m\,(x-y)^k\,(y'-z)^{m'}\,(x'-y')^{k'}}{(y-z)^2\,(x-y)^2\,(y'-z)^2\,(x'-y')^2}\\ \left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^+ + p^+)^2} + \frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k} - \delta_{m'k'}\delta_{km}) + \frac{p^+}{k^{+\,2}}\,\delta_{mm'}\delta_{kk'} + \frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right]\times\\ \left[U^{ce'}(y') - U^{ce'}(x')\right]\times\left[U^{ce}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\end{aligned} \tag{1.3}$$

Using $\xi = \frac{k^+}{(p^+ + k^+)}$ and $1-\xi = \frac{p^+}{(p^+ + k^+)}$, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{N_c}{k^+}\int_{k^+/\vee}^{\frac{k^+}{k^+ + \wedge}}\frac{d\xi}{\xi^2}\int_z e^{-ik\,(y'-y)\,\frac{1}{\xi}}\int_{x,y,y',x'}\Bigg[d_\perp\,\xi\,(1-\xi)\,\frac{(y-z)\cdot(x-y)\,(y'-z)\cdot(x'-y')}{(y-z)^2\,(x-y)^2\,(y'-z)^2\,(x'-y')^2}$$
$$+\left(\frac{\xi}{(1-\xi)}+\frac{(1-\xi)}{\xi}\right)\frac{(y-z)\cdot(y'-z)\,(x-y)\cdot(x'-y')}{(y-z)^2\,(y'-z)^2\,(x-y)^2\,(x'-y')^2}\Bigg]\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\,\rho^{e'}(x')\rho^e(x) \tag{1.4}$$

(The antisymmetric $\frac{2}{k^+}$ channel has been dropped here: the colour factor is z-independent, so it multiplies $\int_z \mathcal{K}^m(y-z)\mathcal{K}^{m'}(y'-z)$, which is symmetric in (m,m'); the antisymmetric contraction vanishes identically at every ξ.)

We are performing the z integral and using $\overline{MS}$ and $d_\perp = 2-\epsilon$, we get

$$\int_z \frac{(y-z)^m\,(y'-z)^{m'}}{(y-z)^2\,(y'-z)^2} = -\frac{\pi}{2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\delta^{mm'} - \pi\,\frac{(y'-y)^m\,(y'-y)^{m'}}{(y'-y)^2} \tag{1.5}$$

This gives us (in the p^+ variables)

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'} e^{-ik\,(y'-y)\,\frac{p^++k^+}{k^+}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{p^+}{(k^++p^+)^2}+\frac{p^+}{k^{+\,2}}+\frac{1}{p^+}\right]\times$$
$$\left[\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\,\rho^{e'}(x')\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'} e^{-ik\,(y'-y)\,\frac{p^++k^+}{k^+}}\left[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,\frac{d_\perp\,p^+}{(k^++p^+)^2}\right.$$
$$\left.+\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{p^+}{k^{+\,2}}+\frac{1}{p^+}\right]\right]\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\,\rho^{e'}(x')\rho^e(x) \tag{1.6}$$

Using $\xi = \frac{k^+}{(p^+ + k^+)}$, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^+ + \wedge}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,y,y',x'} e^{-ik\,(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)}\right]\frac{1}{\xi^2}$$
$$\left[\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\,\rho^{e'}(x')\rho^e(x)$$
$$+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^+ + \wedge}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,y,y',x'} e^{-ik\,(y'-y)\,\frac{1}{\xi}}\,\frac{1}{\xi^2}\left[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\,\xi(1-\xi)\right.$$
$$\left.+\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)}\right]\right]\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\,\rho^{e'}(x')\rho^e(x) \tag{1.7}$$

The cross section is given by

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\times\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]$$
$$\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)_+}\right]\right]\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\,\rho^{e'}(x')\rho^e(x)$$
$$+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\left[\log\frac{(\vee-k^+)\,k^+}{\vee\,\wedge}\right]\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.$$
$$\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\,\rho^{e'}(x')\rho^e(x)$$
$$+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^+ + \wedge}}d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}$$
$$\times\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\,\rho^{e'}(x')\rho^e(x) \tag{1.8}$$

Using $d_\perp = 2-\epsilon$, we get

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& -\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\right]\\
&\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\,\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x)\\
&-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\left[\log\frac{k^+}{\Lambda}+\log\left(1-\frac{k^+}{\vee}\right)\right]\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\\
&\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x)\\
&+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]\\
&\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x) \qquad (1.9)\end{aligned}$$

The JIMWLK term is straightforward and is given by

$$\begin{aligned}\frac{d^3N_{\rm JIMWLK}}{d^3k} =& \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[-\frac{2}{\epsilon}-2\gamma_E-\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\\
&\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x) \qquad (1.10)\end{aligned}$$

This term can be written as using the following relation

$$\left[-\frac{2}{\epsilon}-2\gamma_E-\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]=\frac{1}{\pi}\int_z\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2} \qquad (1.11)$$

$$\begin{aligned}\frac{d^3N_{\rm JIMWLK}}{d^3k} =& \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,z,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\\
&\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x) \qquad (1.12)\end{aligned}$$

Now, let us move on to obtaining the BFKL term. It is obtained by removing the pole at $\xi\to 0$. So let's go back to eq. (**??**) and write the term which has the $\xi\to 0$ pole:

$$\begin{aligned}\mathcal{D} =& -\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\\
&\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x) \qquad (1.13)\end{aligned}$$

Rewriting them using (**??**), we get the following

$$\begin{aligned}\mathcal{D} =& \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,\frac{1-\xi}{\xi}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2}\\
&\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x) \qquad (1.14)\end{aligned}$$

We can rewrite this term as follows:

$$\begin{aligned}\mathcal{D} =& \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,w,w',x',y',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\\
&\delta^{(2)}\left(y-z-\xi\,(w-z)\right)\,\delta^{(2)}\left(y'-z-\xi\,(w'-z)\right)\,\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x) \qquad (1.15)\end{aligned}$$

This term is clearly divergent at $\xi\to 0$ and thus we use the + prescription, we get the following

$$\begin{aligned}\mathcal{D} =& \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,w,w',x',y',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\\
&\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x)\times\Bigg[\delta^{(2)}\left(y-z-\xi\,(w-z)\right)\,\delta^{(2)}\left(y'-z-\xi\,(w'-z)\right)\\
&-\delta^{(2)}(y-z)\,\delta^{(2)}(y'-z)\Bigg]\\
&+\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,w,w',x',y',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\\
&\delta^{(2)}(y-z)\,\delta^{(2)}(y'-z)\,\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x) \qquad (1.16)\end{aligned}$$

We can represent the first term like this:

$$\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,w,w',x',y',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\times\Big[\delta^{(2)}\left(y-z-\xi\,(w-z)\right)\,\delta^{(2)}\left(y'-z-\xi\,(w'-z)\right)$$
$$-\,\delta^{(2)}(y-z)\,\delta^{(2)}(y'-z)\Big]=$$
$$\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\int_{k^+/\vee}^1 d\xi\left[\frac{(1-\xi)}{\xi}\right]_+\int_{x,y,w,w',x',y',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\times\,\delta^{(2)}\left(y-z-\xi\,(w-z)\right)\,\delta^{(2)}\left(y'-z-\xi\,(w'-z)\right)\tag{1.17}$$

Definition of this plus prescription. The "+" here is at $\xi=0$ and is *defined by* eq. (**??**): the subtraction term is the integrand with the δ-functions evaluated at $\xi=0$, i.e. $\delta^{(2)}(y-z)\delta^{(2)}(y'-z)$ (the position collapse $y,y'\to z$), all other factors untouched. It must not be confused with the $(1-\xi)_+$ used elsewhere in this note, which subtracts at $\xi=1$. Wherever $\left[\frac{1-\xi}{\xi}\right]_+$ appears below — in particular inside $\mathcal{P}_{gg,+}$ and under the bracket — it is shorthand for exactly this δ-form subtraction and nothing else; see the remark after eq. (**??**).

Integrating wrt y,y' in the second term, we get

$$\mathcal{D}=\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\int_{k^+/\vee}^1 d\xi\left[\frac{(1-\xi)}{\xi}\right]_+\int_{x,y,w,w',x',y',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\times\,\delta^{(2)}\left(y-z-\xi\,(w-z)\right)\,\delta^{(2)}\left(y'-z-\xi\,(w'-z)\right)$$
$$+\,\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,w,w',x',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}$$
$$\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\tag{1.18}$$

Now integrating the second term wrt ξ, we get

$$\mathcal{D}=\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\int_{k^+/\vee}^1 d\xi\left[\frac{(1-\xi)}{\xi}\right]_+\int_{x,y,w,w',x',y',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\times\,\delta^{(2)}\left(y-z-\xi\,(w-z)\right)\,\delta^{(2)}\left(y'-z-\xi\,(w'-z)\right)$$
$$+\,\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{\vee}{k^+}\right)-\left(1-\frac{k^+}{\vee}\right)\right]\int_{x,w,w',x',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}$$
$$\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\tag{1.19}$$

Integrating the first term wrt w,w' we get

$$\mathcal{D}=\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2}\left[\frac{(1-\xi)}{\xi}\right]_+\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)$$
$$+\,\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{\vee}{k^+}\right)-\left(1-\frac{k^+}{\vee}\right)\right]\int_{x,w,w',x',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}$$
$$\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\tag{1.20}$$

Now, using (**??**) we get

$$\mathcal{D}=-\,\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\,\frac{1}{k^+}\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2}\left[\frac{(1-\xi)}{\xi}\right]_+\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)$$
$$+\,\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{\vee}{k^+}\right)-\left(1-\frac{k^+}{\vee}\right)\right]\int_{x,w,w',x',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}$$
$$\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\tag{1.21}$$

Important: the first term of (??) is shorthand, defined by (??). The identity (**??**) applies to the *unsubtracted* piece of the plus bracket, where the only z-dependence is $K(y,y',z)$. In the *subtraction* piece the z-dependence sits in three factors, $K(w,w',z)\,K(x,x',z)\,[U(z)-U(x')][U(z)-U(x)]$, so (**??**) cannot be used on it — its z-integral is a genuine finite integral (after

the w, w' integration it equals $\frac{4\pi^2}{k^2}\,K(x,x',z)\,W(z,z)$-type structures), *not* $-\pi$. Nor can the "+" in (**??**) be defined intrinsically there: the would-be $\xi \to 0$ limit of $\frac{1}{\xi^2} e^{-ik(y'-y)/\xi}(\cdots)$ does not exist pointwise (the $\log(y'-y)^2$ grows in the collapse region). Therefore $\left[\frac{1-\xi}{\xi}\right]_+ \otimes$ in (**??**), and inside $\mathcal{P}_{gg}$ below, **means**: the unsubtracted piece in the form *minus* the subtraction piece in its δ/z-form,

$$\int_{\bar\xi}^1 \frac{d\xi}{\xi^2}\left[\frac{1-\xi}{\xi}\right]_+(\cdots) \equiv \int_{\bar\xi}^1 \frac{d\xi}{\xi^2}\,\frac{1-\xi}{\xi}\,(\cdots) + \frac{1}{\pi}\int_{\bar\xi}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,w,w',x',z} e^{-ik(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\,W(z,z)$$

(relative + sign because the $-\pi$ of (**??**) has been divided out of the overall prefactor). With this one definition, every later equation of the note is exact.

The BFKL term is:

$$\begin{aligned}\frac{d^3N_{\rm BFKL}}{d^3k} =& \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,z,w,x',w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\ &\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x)\end{aligned} \tag{1.22}$$

The remaining term is as follows:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} =& -\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^4}\,\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\xi(1-\xi)+\left[\frac{1-\xi}{\xi}\right]_+ + \frac{\xi}{(1-\xi)_+}\right]\\ &\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x)\\ &-\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^4}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\\ &\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x)\\ &+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]\\ &\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x)\\ &-\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,w,w',x',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\\ &\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x)\end{aligned} \tag{1.23}$$

Now, let us move on to extracting the DGLAP and running-coupling contribution. In the above equation, we denote

$$\mathcal{P}_{gg,+}(\xi) = 2N_c\left[\xi\,(1-\xi)+\left[\frac{(1-\xi)}{\xi}\right]_+ + \frac{\xi}{(1-\xi)_+}\right]$$

and thus we write the gluon-gluon splitting function ($\mathcal{P}_{gg}$) as follows:

$$\mathcal{P}_{gg}(\xi) = \mathcal{P}_{gg,+}(\xi) + \frac{\beta_0}{2}\,\delta(1-\xi) \tag{1.24}$$

Caution on this $\mathcal{P}_{gg}$. This is *not* the textbook splitting function: the textbook $\mathcal{P}_{gg,+}$ has an *unregulated* $\frac{1-\xi}{\xi}$, while here that channel carries the $\xi=0$ subtraction of (**??**). The difference between the two is exactly the BFKL subtraction (the pole term that produced $\log\frac{\vee}{k^+}$ above), so this "BFKL-subtracted" splitting function is the natural object in the present scheme — but it should be distinguished notationally (e.g. $\widetilde{\mathcal{P}}_{gg}$) when comparing with the DGLAP literature. The relation to $\beta_0 = \frac{11N_c}{3}$ via the $\delta(1-\xi)$ term is unaffected (it concerns the $\xi=1$ endpoint only). Also recall that the two plus prescriptions in this $\mathcal{P}_{gg,+}$ act at *opposite* endpoints: $\left[\frac{1-\xi}{\xi}\right]_+$ at $\xi=0$ (defined by (**??**)), $\frac{\xi}{(1-\xi)_+}$ at $\xi=1$ (locally subtracted on $[\frac{k^+}{\vee},1]$).

So, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^4}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\mathcal{P}_{gg}(\xi)-\frac{\beta_0}{2}\delta(1-\xi)\right]$$
$$\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,w,w',x',z} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}$$
$$\left[U^{ce'}(z)-U^{ce'}(x')\right]\left[U^{ce}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.25}$$

In the first term, integrating with respect to ξ for the β_0 term (the $\delta(1-\xi)$ sits at $\xi=1$ inside the range, where $\frac{1}{\xi^2}=1$ and the phase is the LO phase; no limit $\vee\to\infty$ is needed), we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2}{8\pi^2}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\mathcal{P}_{gg}(\xi)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left(\frac{2}{\epsilon}+2\,\gamma_E+\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\times$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$+\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2\,\beta_0}{16\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left(\frac{2}{\epsilon}+2\,\gamma_E+\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\times$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,w,w',x',z} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}$$
$$\left[U^{ce'}(z)-U^{ce'}(x')\right]\left[U^{ce}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.26}$$

violet - DGLAP and green - running coupling. We write the final result in different categories:

1.1 JIMWLK

$$\frac{d^3N_{\rm JIMWLK}}{d^3k} = +\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,z,x',y'} e^{-ik\cdot(y'-y)}\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.27}$$

(Sign corrected to +: this category must be identical to (??), which carries + — the bracket $[-\frac{2}{\epsilon}-2\gamma_E-\log]$ of (??) equals $+\frac{1}{\pi}\int_z K$ by (??), with no extra minus sign.)

1.2 BFKL

$$\frac{d^3N_{\rm BFKL}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{x,z,w,x',w'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\left[U^{ce'}(z)-U^{ce'}(x')\right]\left[U^{ce}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{1.28}$$

1.3 DGLAP

$$\frac{d^3N_{\rm DGLAP}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2}{8\pi^2}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\mathcal{P}_{gg}(\xi)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left(\frac{2}{\epsilon}+2\,\gamma_E+\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\times$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x) \tag{1.29}$$

(with $\mathcal{P}_{gg}$ the BFKL-subtracted splitting function defined above; the $\left[\frac{1-\xi}{\xi}\right]_+$ channel is shorthand for the pair of (??)/(??).)

1.4 Running coupling

$$\frac{d^3N_{\rm RC}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2\,\beta_0}{16\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left(\frac{2}{\epsilon}+2\,\gamma_E+\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\times$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x) \tag{1.30}$$

1.5 Finite terms

$$\begin{aligned}\frac{d^3N_{\rm Finite}}{d^3k} =& -\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,z,w,x',w'}e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\
&\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\left[U^{ce}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\\
&-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\\
&\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\\
&+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]\\
&\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\end{aligned} \tag{1.31}$$

Remark on the second entry. The $\log(1-\frac{k^+}{\vee})$ term carries $\frac{2}{\epsilon}$, so this block is finite only up to the plus-prescription convention: that entry is exactly the difference between the locally subtracted $\frac{\xi}{(1-\xi)_+}$ on $[\frac{k^+}{\vee},1]$ used in $\mathcal{P}_{gg,+}$ above and the standard $[0,1]$ plus distribution, $\int_{\bar\xi}^1 d\xi\,\frac{h(\xi)}{(1-\xi)_+}=\int_{\bar\xi}^1 d\xi\,\frac{h(\xi)-h(1)}{1-\xi}+h(1)\log(1-\bar\xi)$. If the standard convention is adopted for the $\xi=1$ plus prescription, this entry is absorbed into the DGLAP term (where its pole joins the collinear kernel) and the finite block — then just the boundary and azimuthal entries — is free of $\frac{1}{\epsilon}$, μ, $\log\Lambda$ and large $\log\frac{\vee}{k^+}$. Its coefficient $\log(1-\frac{k^+}{\vee})$ is in any case bounded (no large logarithm).

2 Finiteness of the last (bracket) term: momentum-space derivation

(Added to settle two questions about the last entry of the finite terms: whether it diverges at $y=y'$, and whether its ξ-integral hides a log $\frac{\vee}{k^+}$. The answer to both is no, and in momentum space both statements can be read off from a single Fourier transform.)

The term in question is

$$\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\,A(\xi)\,,\qquad A(\xi)=\int_{x,y,x',y'}e^{-ik\cdot r\,\frac{1}{\xi}}\,\mathcal{K}^i(x-y)\,\mathcal{K}^j(x'-y')\,t^{ij}(r)\,W(y,y')\,, \tag{2.1}$$

with $r=y'-y$, $\mathcal{K}^i(u)=\frac{u^i}{u^2}$, $W(y,y')$ the Wilson-line/colour factor, and

$$t^{ij}(r)\;=\;\delta^{ij}-2\,\hat r^i\hat r^j\,,\qquad \hat r=\frac{r}{|r|}\,, \tag{2.2}$$

which is the square bracket of (??): the apparent $\frac{1}{(y'-y)^2}$ has already cancelled against the two factors of $(y'-y)$ in the numerator, so only the *direction* $\hat r$ survives — t^{ij} is bounded, $|t^{ij}|\le\ldots$, everywhere, including $r\to 0$.

Step 1: t^{ij} is traceless in two dimensions. $\delta^{ii}=2$ and $2\,\hat r\cdot\hat r=2$, so $t^{ii}=0$. Consequently t^{ij} has no isotropic part: its angular average vanishes, $\int_0^{2\pi}d\theta_{\hat r}\,t^{ij}(\hat r)=0$.

Step 2: the Fourier transform of t^{ij}. The angular integral couples only to the second angular Fourier mode of the plane wave, $\int_0^{2\pi}d\theta_{\hat r}\,t^{ij}(\hat r)\,e^{iq\cdot r}=2\pi J_2(q|r|)\left(2\hat q^i\hat q^j-\delta^{ij}\right)$, and the radial integral has the closed form $\int_0^Z z\,J_2(z)\,dz=2-2J_0(Z)-Z\,J_1(Z)$, whose distributional value (what any smooth test function sees) is 2. Hence

$$\boxed{\widetilde{T}^{ij}(q)=\int d^2r\;e^{iq\cdot r}\left[\delta^{ij}-2\hat r^i\hat r^j\right]=\frac{4\pi}{q^2}\left(2\hat q^i\hat q^j-\delta^{ij}\right)} \tag{2.3}$$

No $\delta^{(2)}(q)$ (Step 1), no constant, no $\log q$ — a decaying power, times a traceless angular factor. (Verified distributionally: smearing both sides of (**??**) against a Gaussian test function centred at $q_0 = (1.7, 0)$ gives $11.573073 = 11.573073$ for the xx component, $-11.573073 = -11.573073$ for yy, $0 = 0$ for xy.)

Step 3: the term as a momentum-space convolution. Writing $y' = y + r$ and Fourier-transforming the r-dependence,

$$A(\xi) = \frac{1}{(2\pi)^2} \int d^2q\, \widetilde{T}^{ij}(q)\, \Phi^{ij}\Big(\frac{k}{\xi} - q\Big)\,, \qquad \Phi^{ij}(P) = \int_{x,x',y} \int d^2r\, e^{-iP\cdot r}\, \mathcal{K}^i(x-y)\, \mathcal{K}^j(x'-y-r)\, W(y, y+r)\,, \tag{2.4}$$

where Φ^{ij} is the transform of everything smooth (kernels away from the coincidence point, Wilson lines); it is finite and decays at large $|P|$.

Step 4: finiteness, read off from (??).

(a) *The point $y = y'$ = the large-q region.* A genuine short-distance singularity would appear as a non-decaying transform: $\mathrm{FT}[\delta^{(2)}(r)] = 1$ (constant), $\mathrm{FT}[1/r^2] \sim -2\pi \log q$ (growing). Here instead $\widetilde{T} \sim \frac{1}{q^2}$ *decays*, so the q-integral of (**??**) converges absolutely at large q: the term has no divergence at coincident points, lives in exactly two transverse dimensions, and needs no regulator and no μ.

(b) *The $q \to 0$ end.* $\frac{1}{q^2}\, d^2q$ is marginal, but the traceless angular factor rescues it: $\int d\theta_{\hat{q}} (2\hat{q}^i\hat{q}^j - \delta^{ij}) = 0$, so the leading term of $\Phi(\frac{k}{\xi} - q)$ around the regular point $q = 0$ drops and the remainder is $\int d^2q\, \frac{O(q)}{q^2}$, convergent.

(c) *The $\xi \to 0$ end = the $\log \vee$ question.* As $\xi \to 0$ the convolution is dominated by $q \approx \frac{k}{\xi}$, so

$$A(\xi) \;\xrightarrow[\xi\to 0]{}\; \widetilde{T}^{ij}\Big(\frac{k}{\xi}\Big)\, \Phi^{ij}(0) + \ldots = 4\pi\, \frac{\xi^2}{k^2}\, \big(2\hat{k}^i\hat{k}^j - \delta^{ij}\big)\, \Phi^{ij}(0) + \ldots = O(\xi^2)\ : \tag{2.5}$$

the $\frac{1}{q^2}$ of (**??**), evaluated at the large momentum $\frac{k}{\xi}$, is where the ξ^2 comes from. Therefore $\int_{k^+/\vee}^1 d\xi\, \frac{1-\xi}{\xi}\, A(\xi) = \int d\xi\, (1-\xi)\, \xi \times (\text{const}) + \ldots$ converges: **no** $\log \frac{\vee}{k^+}$, and the dependence on the lower limit is a power correction $O\big((k^+/\vee)^2\big)$, so the limit $\vee \to \infty$ exists.

Step 5: quantitative check. Step 4(c) is a parameter-free prediction. In a model with the exact bracket structure ($\mathcal{K}_1 = (0.7, -1.2)$, $\mathcal{K}_2 = (0.4, 2.1)$ frozen at the coincidence point, $k = (2, 0)$, a Gaussian for W), the direct two-dimensional position-space integration gives $A(\xi)/\xi^2 = 8.796$, constant to five digits from $\xi = 0.4$ down to $\xi = 0.1$; the momentum-space formula predicts

$$\frac{A(\xi)}{\xi^2} \;\to\; \frac{4\pi}{k^2}\Big[2(\mathcal{K}_1\cdot\hat{k})(\mathcal{K}_2\cdot\hat{k}) - \mathcal{K}_1\cdot\mathcal{K}_2\Big] = \frac{4\pi}{4}\big[2(0.7)(0.4) - (0.28 - 2.52)\big] = 2.8\,\pi = 8.7964594\,, \tag{2.6}$$

in exact agreement. Together with the elementary position-space facts (the bracket is bounded at $y = y'$ by $\frac{3}{|x-y||x'-y|}$, with a direction-dependent but harmless value at the single point $y' = y$), this establishes that the last term of the finite block is finite in every sense: no singularity at coincident points, no ultraviolet regulator needed, and no large longitudinal logarithm from the ξ-integration.